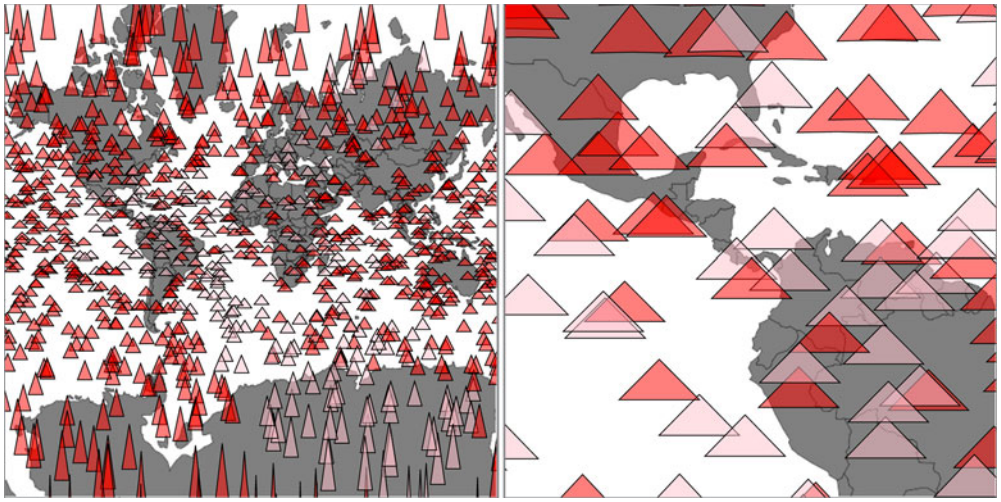


Figure 11.5 Background countries are drawn with canvas, while foreground triangles are drawn with SVG to use interactivity. SVG graphics are individual elements in the DOM and therefore amenable to having click, mouseover, and other event listeners attached to them.

being zoomed and panned, but render them in SVG when the map isn't in motion and the user is mousing over certain elements.

How do you determine when a zoom event starts and when it finishes? In the past you had to set a timer, check to see if the user was still zooming, and then redraw the elements. But, fortunately, D3 introduced a pair of new events to the zoom control: `"zoomstart"` and `"zoomend"`. These fire, as you may guess, when the zoom event begins and ends, respectively. The following listing shows how you'd initialize a zoom behavior with different functions for these different events.

Listing 11.8 Mixed rendering based on zoom interaction

```
var projection = d3.geo.mercator().scale(100).translate([250,250]);
var svgPath = d3.geo.path().projection(projection);
var canvasPath = d3.geo.path().projection(projection);

mapZoom = d3.behavior.zoom()
    .translate(projection.translate())
    .scale(projection.scale())
    .on("zoom", zoomed)
    .on("zoomstart", zoomInitialized)
    .on("zoomend", zoomFinished);

d3.select("svg").call(mapZoom);

var g = d3.select("svg").append("g")

g.selectAll("path.sample").data(sampleData)
    .enter()
    .append("path")
```

Assigns separate functions for each zoom state